

VIEWPOINT

HEALTH CARE POLICY AND LAW

Body Composition in Anti-Obesity Medication Trials—Beyond Scales

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Obesity is the world's most prevalent chronic disease and has increasingly been recognized as a global public health challenge. It has alarming health effects and burdens health care systems globally. From 2013 to 2016, approximately half of US adults undertook weight loss initiatives, reflecting widespread concerns about weight and efforts toward healthier living. Concurrently, the ascent of anti-obesity medications (AOMs) into mainstream consciousness is attributable to their reported efficacy and the scarcity of cost-effective or accessible alternatives for a vast proportion of the population.¹ Crafting appropriate interventions for obesity, however, mandates the creation of proper clinical end points. We argue that, by focusing only on weight loss as the primary weight medication end point, the US Food and Drug Administration (FDA) is encouraging inaccurate measures of medication efficacy for both patients and clinicians.

Current FDA Processes

The FDA's criteria for weight loss medications require that therapeutics show a minimum 5% weight loss over the placebo or that 35% of participants achieve a 5% weight loss, doubling the placebo group's rate.² Currently, 7 medications, including the recently approved tirzepatide, are FDA approved for long-term weight management.³ Of note, the available FDA releases for these medications report on body mass or body mass index (BMI) changes without mention of fat reduction.⁴

However, equating weight loss with improved health outcomes is misleading to clinicians and consumers, as fat loss is a much greater factor in health consequences than overall weight loss.⁵ It is crucial to note that visceral fat is specifically associated with metabolic complications, such as insulin resistance, type 2 diabetes, heart disease, and certain cancers.⁶ If FDA metrics prioritize overall weight loss without differentiating the source of that loss, medications may be deemed effective or superior to others based on weight reduction alone despite not necessarily achieving greater fat loss, which is a more relevant measure of health outcomes. If AOMs are indeed designed to treat obesity, it also merits attention that the American Medical Association recently decided to deviate from weight alone as a metric for determining obesity, with international standards soon following suit.

Overall weight loss, including via AOMs, occurs via both fat and muscle mass loss; the health outcomes of a patient are determined by the proportion of these 2 values. Excessive loss of lean muscle mass can cause sarcopenic obesity, which is associated with functional decline and increased mortality.⁷ As such, while vis-

ceral fat loss is beneficial, it should be accompanied by preservation of or an increase in muscle mass for ideal health outcomes. More research is needed to clarify the complex relationships between different types of weight loss, body composition, and health outcomes.

The pharmaceutical industry's significant role in developing and marketing AOMs must be balanced against the potential health and economic consequences of drug approvals based on insufficient criteria. Especially for vulnerable groups like Medicare patients, who are more prone to obesity and muscle loss, the FDA's role in providing guidelines for drug efficacy becomes critical. If AOMs miss intended results or cause unintended consequences by reducing lean mass more than fat, they risk increasing illness and medical costs, significantly straining an already burdened system. Thus, if FDA guidelines are not stringent enough, the resultant financial burdens will be felt by all.

Discrepancies between the FDA's approval mechanisms and the actual outcomes of AOMs underscore the need for a more holistic framework. Such a reformed system could set new standards for health care globally, ensuring better patient care.

Measuring What Matters

To encourage weight loss that optimizes health, the FDA should require fat loss, rather than weight loss, as the primary end point of AOM trials. Establishing a baseline fat level for participants and monitoring body fluctuations would ensure that weight loss comes from fat, preferably visceral fat, and would unveil any adverse effects, including undesirable muscle loss.⁸ This method can pave the way for more tailored treatment plans and stringent approval benchmarks. Most important, by targeting fat loss, this method would address the primary cause of weight-related health conditions.

To analyze body composition equitably and accurately, we recommend incorporating accessible analytical methods alongside biomedical indices (eg, low- and high-density lipoprotein levels) in drug trials. Such methods include dual-energy x-ray absorptiometry (DXA), magnetic resonance imaging, computed tomography, bioelectrical impedance analysis, and visual body composition (VBC).⁹ Although DXA, magnetic resonance imaging, and computed tomography are criterion standards, they are costly, often operator dependent, and difficult to scalably administer.^{8,9} For instance, DXA accessibility is constrained for individuals with higher BMI due to the equipment's weight limits. Visual body composition has recently emerged as a more viable alternative, leveraging conventional smartphone cameras to estimate total body fat percentage.⁹ In a recent study,⁹

VBC had good concordance with DXA for body fat percentage measurement and was accurate and less biased compared with DXA references. We therefore suggest affordable, scalable methods such as bioelectrical impedance analysis and VBC, which are cost-effective for extensive trials. Alongside biomedical indices, this approach will provide a clear picture of patient health improvement.

A potential downside to this approach is that body composition measurements could inadvertently slow approval processes for upcoming therapeutics. This may also hold AOM trials to a higher standard than other chronic disease treatments, biasing against innovation for patients struggling with obesity. Thus, it is crucial to strike a balance, ensuring rigorous standards without hampering the speed of critical advancements in creating or disseminating medical therapeutics.

To adhere to the broad mission of AOMs, which is to improve obesity-related outcomes such as cardiovascular disease and cancer, it is essential that we measure these outcomes in patients with obesity. However, it is uncertain whether fat loss can serve as a surrogate outcome for these diseases. Although fat reduction can improve health markers linked to reduced risk of chronic illnesses, other health elements like muscle mass, fitness, diet, and lifestyle are also vital. Further research is necessary to evaluate whether fat loss reliably predicts outcomes for obesity-related diseases.

The Role of Clinicians

Beyond the FDA's oversight on the safe use of AOMs, the onus is on physicians to provide comprehensive guidance to their patients. Media outlets significantly shape public opinion, sometimes promoting well-informed decisions and other times fostering misconceptions about weight loss treatments. Clinicians must counter these influences by making patients aware of potential adverse effects associated with AOMs, such as gastrointestinal disorders (eg, vomiting, diarrhea) and abdominal pain, and strategies to mitigate them.⁴ Patients must also understand that AOMs will only achieve and sustain their intended effects with concurrent lifestyle interventions. For example, patients with lower BMIs who neglect exercise often exhibit worse cardiovascular disease risk profiles than those with higher BMIs who are physically active.¹⁰ Last, it is key that physicians make efforts to help patients promote lean muscle retention or growth. Such transformative weight loss conversations rely on a supportive, collaborative approach between clinicians and patients.

In the evolving landscape of weight loss therapeutics, the role of the medical community and regulatory bodies like the FDA must be to ensure individuals' health, well-being, and safety in their quest for healthier lives. We call for transparency, responsibility, and, most important, prioritization of genuine health outcomes over metrics derived from convenience rather than clinical efficacy, ensuring patient safety and the best possible results in the fight against obesity.

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REFERENCES

1. Wing RR. The challenge of defining the optimal lifestyle weight loss intervention for real-world settings. *JAMA*. 2022;328(22):2213-2214. doi:10.1001/jama.2022.21908
2. Morrato EH, Allison DB. FDA approval of obesity drugs: a difference in risk-benefit perceptions. *JAMA*. 2012;308(11):1097-1098. doi:10.1001/jama.2012.10007
3. Idrees Z, Cancarevic I, Huang L. FDA-approved pharmacotherapy for weight loss over the last decade. *Cureus*. 2022;14(9):e29262. doi:10.7759/cureus.29262
4. FDA approves new drug treatment for chronic weight management, first since 2014. US Food & Drug Administration. June 4, 2021. Accessed October 18, 2023. <https://www.fda.gov/news-events/press-announcements/fda-approves-new-drug-treatment-chronic-weight-management-first-2014>
5. Xiang J, Ding XY, Zhang W, et al. Clinical effectiveness of semaglutide on weight loss, body composition, and muscle strength in Chinese adults. *Eur Rev Med Pharmacol Sci*. 2023;27(20):9908-9915. doi:10.26355/eurrev.202310_34169
6. Goodpaster BH, Kelley DE, Wing RR, Meier A, Thaete FL. Effects of weight loss on regional fat distribution and insulin sensitivity in obesity. *Diabetes*. 1999;48(4):839-847. doi:10.2337/diabetes.48.4.839
7. Batsis JA, Villareal DT. Sarcopenic obesity in older adults: aetiology, epidemiology and treatment strategies. *Nat Rev Endocrinol*. 2018;14:513-537. doi:10.1038/s41574-018-0062-9
8. Ponti F, Plazzi A, Guglielmi G, Marchesini G, Bazzocchi A. Body composition, dual-energy X-ray absorptiometry and obesity: the paradigm of fat (re)distribution. *BJR Case Rep*. 2019;5(3):20170078. doi:10.1259/bjrcr.20170078
9. Majmudar MD, Chandra S, Yakkala K, et al. Smartphone camera based assessment of adiposity: a validation study. *NPJ Digit Med*. Published online June 29, 2022. doi:10.1038/s41746-022-00628-3
10. Wei M, Kampert JB, Barlow CE, et al. Relationship between low cardiorespiratory fitness and mortality in normal-weight, overweight, and obese men. *JAMA*. 1999;282(16):1547-1553. doi:10.1001/jama.282.16.1547